

Facilitating Social Initiations of Preschoolers With Autism Spectrum Disorders Using Video Self-Modeling

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Abstract

Video self-modeling (VSM) has accumulated a relatively impressive track record in the research literature across behaviors, ages, and types of disabilities. Using only positive imagery, VSM gives individuals the opportunity to view themselves performing a task just beyond their present functioning level via creative editing of videos using VCRs or video software. In this study, the authors used a single-subject multiple-baseline design across four preschool children on the autism spectrum to determine if self-modeling would facilitate their social initiations during playground time. The results were mostly positive, with two children exhibiting major treatment effects, one with questionable results, and one child being unaffected. Discussion of the results, the practicality of VSM, and suggestions for continued research are provided.

Keywords

autism spectrum disorders, social initiations, video self-modeling

For many years, parents, teachers, and therapists have actively pursued a wide range of approaches to treating and teaching individuals with autism spectrum disorders (ASD). The enigmatic nature of behaviors associated with ASD, especially those related to language, social skills, and attending to tasks, has been a source of frustration for families and professionals. At present, the state of the art in treatment remains confusing. Controversy surrounds many forms of intervention due to the obtrusiveness of intervention such as discrete trial training with younger children with ASD (Smith, 2001) or the lack of, or conflicting, supporting evidence, as is the case of facilitated communication (Mostert, 2001).

A relatively new genre of treatments that holds promise for individuals with ASD has focused on the use of videos. Although video modeling is being shown to be an effective instructional procedure for individuals with ASD (e.g., D'Ateno, Mangiapanello, & Taylor, 2003; Hine & Wolery, 2006; Paterson, Arco, & Cowan, 2007), there may be a form of this technique that is even more effective. Video-taping individuals acting as their own models has been proposed as a method that promotes attention to the model (Creer & Miklich, 1970; Hosford, 1981). Researchers have found that the most effective models tend to be individuals close to the observer's age, with similar characteristics (gender, personality, race, and mood), and who are functioning only slightly above the level of the observer (Thoresen & Hosford, 1973). These factors related to similarities of

observer and model lead to the question of whether individuals could serve as their own best models if given the opportunity. No peer or adult can exhibit characteristics as close and relevant to individuals as they can themselves. Bandura (2001) talks about the concept of self-efficacy, the personal belief that one can succeed at a task. He believes, and his research supports, that this is a critical factor in learning and performance. Video self-modeling (VSM) may serve to promote self-efficacy because it provides the viewer with visual evidence that s/he can accomplish the task.

For years, the technology necessary for editing videos was complex and costly, probably serving as a barrier to applying VSM in homes, clinics, and classrooms. However, in recent years both Apple® and personal computers have been equipped with free software (iMovie® and Movie Maker®, respectively) that makes editing video as easy as using a word processor. Now the technology is available to allow observers to act as their own models. In fact, effective self-modeling videos can be produced simply by hooking up a camcorder to a VCR or DVD player and extracting desired video clips (see Buggy, 2007; Dowrick, 1983).

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VSM in which observers are shown only their positive performances of a targeted behavior has been shown to be an effective treatment across an extensive range of behaviors, ages, and abilities. Positive results have been obtained for treating depression (Kahn, Kehle, Jenson, & Clark, 1990), stuttering (Bray & Kehle, 1996), elective mutism (Pigott & Gonzales, 1987), attention disorders (Dowrick & Raeburn, 1977; Woltersdorf, 1992), behavior disorders (Lasater & Brady, 1995), and aggressive behaviors (Buggey, 2007; Creer & Miklich, 1970; McCurdy & Shapiro, 1988). Likewise, VSM has proven to be efficacious as a tool for teaching skills such as math (Schunk & Hanson, 1989), life skills (Miklich, Chida & Danker-Brown, 1977), social behaviors (Bellini, Akullian, & Hopf, 2007; Loncke, Brady, McPherson, & Hawkins, 1994; Rasing, Coninx, Duker, & Van Den Hurk, 1994), and language (Buggey, 1995, 2005; Haarmann & Greelis, 1982; Sherer et al., 2001; Wert & Neisworth, 2003; Whitlow & Buggey, 2003). In a review of 27 studies using VSM (Meharg & Woltersdorf, 1990), moderate to dramatic gains were reported across all target behaviors. In another review (Hitchcock, Dowrick, & Prater, 2003), it was found that just over 200 subjects had participated in published research in which VSM was the independent variable.

The number of studies using video modeling in all forms (adult, peer, self, and animation) with individuals with ASD is small, although growing. Specifically targeting studies with children with autism, Delano (2007b) reviewed all forms of video-modeling methods, including adult, peer, and self. She found 19 such studies, 5 of which dealt with self-modeling. Ayres and Langone (2005) conducted a review of all studies dealing with video instruction with students with ASD. This included 14 studies using some form of video modeling, but only 4 with self-as-model. Bellini and Akullian (2007) took the review of literature one step further by conducting a meta-analysis of studies using video modeling with children with ASD. They found 23 such studies, including 8 that evaluated VSM use with individuals with autism. Findings across studies seemed to indicate that acquisition of new skills was greatly facilitated, the generalization of taught skills was occurring, and the maintenance of skills was strong. The authors concluded that both peer video modeling and VSM met the Council for Exceptional Children guidelines for research-based practices.

The one exception to the trend in positive results was a study with preschoolers with autism (Clark et al., 1993). A complex reversal design was used to compare VSM with peer modeling. No change was evident in the reduction of aggressive behaviors of six preschoolers following VSM intervention. Other researchers have had more success using VSM with preschoolers and others with autism.

Wert and Neisworth (2003) used VSM to train spontaneous requesting in four preschoolers with autism. The children

were trained to request items via a discrete trial method; however, the resulting requests were rote and there was no generalization to spontaneous requesting (i.e., the children only requested when prompted to do so). The prompted requests were included in the VSM tapes with the prompts and any negative behaviors edited out. Results, collected using a multiple-baseline design across individuals, indicated that all four participants showed substantial gains. The gains in mean production of spontaneous requests ranged from 800% to 1,200%.

In a study involving three adolescents with moderate to severe autism, Buggey, Toombs, Gardner, and Cervetti (1999) focused on teaching verbal responding. During the play sessions conducted in the homes of the participants, a list of 10 questions were asked related to the play. Responses were rare, but with the play sessions occurring 2 nights a week for over a month, enough were collected to make 2-min videos. All of the children increased their responses after watching their videos. Students maintained gains for at least 1 month after the videos were withdrawn. The authors stated that the time commitment for capturing rare behaviors might make this form of VSM too cumbersome for teachers and therapists.

Along with deficits in language, children with ASD often exhibit deficits in social skills. In a recent study, Bellini et al. (2007) examined the effects that VSM had on the social engagement of two 4-year-olds with ASD. The authors stressed the importance of obtaining data in natural environments and accordingly recorded percentage of time engaged with peers during normal playtimes in the classroom. The videos were constructed using footage from brief interactive periods captured in the classroom. They found that after viewing their videos one child's social engagement increased from 3% to 43% of the playtime, and the other child's increased from 6% to 24%. As in other studies, a strong maintenance effect was seen following withdrawal of VSM.

Buggey (2005) reported three studies in one article with children with ASD, using multiple-baseline designs across behaviors and children. In one study, two second graders role-played proper behavior in situations that would normally trigger tantrums. The role-playing was recorded and edited to show appropriate, alternative responses to the triggers. The duration of the tantrums dropped from more than 20 min per day to 0 min after only three viewings. The day following viewing, two episodes where the children exhibited the beginning auras of tantrums were stopped without adult prompting. Maintenance was tracked for 6 months, and no further tantrums were seen in school. In the same article, Buggey reported a study with two boys, ages 10 and 11 years, with mild autism, focusing on social interactions. Once again, role-playing was used to capture the students interacting with peers and staff at the school. Participants

were fed lines from a script, and the best of the “takes” were placed into a video. Both students increased their interactions with peers and adults, and once again, strong maintenance was reported. In the final study reported by Buggey, one 4-year-old participated in a study involving a multiple-baseline design across three behaviors that included pushing, responding to questions, and verbal initiations. A decrease in pushing to almost zero occurred, and gains were reported for both verbal behaviors.

In one of the few studies addressing academic skills, Delano (2007a) used VSM to improve written language skills with three students with Asperger syndrome, ages 13, 15, and 17 years. Specifically, the writing rate or fluency and the number of elements contained in an essay were measured. Students participated in a 30-min training session for word writing and a 60-min session for essay writing. Then they were asked to create a self-modeling video based on a self-monitoring procedure discussed in the training. This included a written script of the monitoring procedure that the student read aloud and footage of the student completing a chart for tracking progress. A multiple-baseline design across responses was used to evaluate results. It was found that both the number of words and number of essay elements increased for all students. All students maintained the word-writing gains as measured on follow-up probes administered 1 week and then 3 months following withdrawal of the videos. Maintenance for the essay elements, however, was not noted.

The limited number of studies that had children with ASD as participants underscores the need to further investigate the impact VSM has on these children. In particular, the mixed results obtained with preschoolers with autism emphasize the need for more research with this age group.

The purpose of this study was to investigate whether VSM could be used to increase the number of social initiations by children with ASD during playground activities. Although not meant as a replication, the present study aligns closely with the work of Bellini, Akullian, and Hopf (2007) in terms of participants and genre of dependent variable. Whereas Bellini et al. used VSM to promote extending the time children spent engaged in play indoors, in the present study the focus was on the children's ability to initiate social interactions on the playground. It is hoped that this study will add to our understanding of the use of VSM with preschoolers with autism in the important area of social skill training.

Method

Participants

Four students who were diagnosed with pervasive developmental disorders—not otherwise specified and who attended a private inclusive preschool in a small urban area participated in the study. Much of the diagnostic information on

the children was obtained close to their third birthdays as they transitioned from early intervention to preschool services. Certified school psychologists from the local public school system conducted the assessments. The assessment instruments varied across the participants, making comparisons problematic. However, the *Childhood Autism Rating Scale* was completed for all four participants during the study. Lucy's mother and teacher completed her rating, school personnel completed the rating for Tim, John's parents and a teacher completed his rating, and Helen's mother and teacher rated her. Information on each child is presented in Table 1.

Lucy. In addition to the assessments identified in Table 1, the *Autism Diagnostic Observation Scale* was administered, and Lucy scored within the autism range for communication and socialization skills. On the playground, Lucy tended to play with vegetation. She found flowers and clover to pick and spent long periods touching and observing foliage on bushes and vines. Occasionally she would swing, slide, and show her flowers to others when prompted. She also seemed to enjoy playing in the sand area and with the mulch that covered the playground.

Tim. Tim's main behavior on the playground was to ride around in a little car that had an enclosed roof and required entrance through a door that opened and closed. The car was foot propelled, and Tim tended to ride in a tight area of about 150 square feet. He would not wear the required helmet, and when approached by an adult to do so, he would abandon the vehicle. At other times, Tim stayed close to the entrance to the school. He did not slide or swing but would engage with the manipulative devices on two of the large pieces of play equipment.

John. John was the youngest of the participants. During the study, his parents reported that he was experiencing eating problems and had developed several stomach ulcers. On the playground, John typically engaged in one of two behaviors. The marquee of a convention center was visible across the street from the playground, as was an outside elevator encased in glass at an adjoining hotel. John often sat on the piece of equipment closest to these features and stared at one or the other. Even when he moved to other equipment, he could be seen looking toward the hotel. His other behavior was playing with balls and watching them roll downward. He would take one, two, or three balls the size of volleyballs onto equipment and roll them down inclines. There were times that John roamed the playground and used the slide. Most of these occurrences were teacher prompted.

Helen. The behavior Helen engaged in the most on the playground was swinging on a tire swing, often with two fellow students whom she typically ignored. She always smiled while swinging, but typically looked either right or left, away from the peers. At other times, she roamed the playground or stayed close to the entrance to the school.

Table 1. Participant Demographics

	Lucy	Tim	John	Helen
Age	4 years 2 months	4 years 2 months	3 years 10 months	4 years 2 months
Diagnosis	PDD-NOS	PDD-NOS	PDD-NOS	PDD-NOS
Childhood Autism Rating Scale				
Total score/conclusion	36/moderate autism	39/severe autism	37.5/severe autism	39/severe autism
Vineland Adaptive Behavior				
Scales (percentile scores)				
Communication	2	—	—	0.1
Daily living	1	—	—	0.2
Socialization	6	—	—	0.1
Motor skills	1	—	—	1
Composite	1	—	—	
Gilliam Autism Rating Scale—2nd				
ed. (standard scores)				
Stereotyped behavior	—	—	9	14
Communication	—	—	6	8
Social interaction	—	—	11	9
Developmental			11	10
Autism quotient/conclusion	—	—	98/autism very likely	95/average autism likelihood
Adaptive Behavior				
Assessment System				
Conceptual	—	—	0.2	—
Social	—	—	0.6	—
Practical	—	—	0.9	—

Note: Dashes indicate that assessments were not conducted. PDD-NOS = pervasive developmental disorders—not otherwise specified.

The four children in this study participated in a different study during half of the previous school year. A buddy system, pairing children with disabilities with peers without disabilities for transitions and center time, along with social simulations dealing with sharing, helping, and respecting individual differences, was carried out with all children in the classroom. The results indicated that social interactions among children with disabilities and peers more than doubled during playground time. However, for the four students in the present study, there was no change in their social initiations, nor was there a change in peer initiations toward them. It was for this reason that another approach (VSM) was selected to facilitate social initiations. Prior to the outset of the study, approval was obtained from the university's institutional review board and the research committee at the institute where the study was conducted. Informed consent was obtained from the parents of the participants, and amendments were added to the children's individualized education programs regarding the treatment.

A simple test was devised to ascertain whether the children could self-recognize. A relatively large view screen on a camcorder was turned so that it was facing the children. Their reactions to their images were noted. All of the children attended well to their images. Three of the four children engaged in making faces or sticking out their tongues. This

display of cause-effect behavior aimed at the camera was considered a good indicator of self-recognition. The other child, Helen, merely smiled at herself, turning and nodding her head as she watched. The researchers felt relatively confident that Helen's behavior was "posing" for the camera, thus also indicating a cause-effect relationship, with her movements resulting in changes in the image in the viewfinder.

Setting

The preschool was part of a larger institute that provided observational settings and practicum placements for several local colleges. The common presence of observers made the children somewhat immune to observer effects, making data collection less obtrusive. The school had an equal number of children with developmental delays and typically developing students. The classrooms were arranged by age, ranging from 6 weeks to 1 year in the youngest classroom to children 4- to 5-plus years in the oldest. The four participants were in classrooms for children ranging in age from 3.6 to 4.6 years. There were two of these classrooms, each with 12 children. Two participants were in each class. The classes were staffed by a lead teacher, two aides, and two additional aides assigned to individual children (not the participants), one in each room. Both teachers had degrees in early childhood education, and the assistants either had

bachelor's degrees or early childhood assistant certification from a community college.

Students received a full range of support services. All of the participants received speech and occupational therapy that were integrated into classroom routines. The classrooms contained low mobile storage shelving serving as dividers for centers. Centers varied in content and revolved around a theme that was common across both classrooms. The playground area was approximately 10,000 square feet, with modern equipment being beta tested at the site. Three large structures with elements for climbing, sliding, and manipulation served as the centerpieces, with swing and sand areas adjoining.

Materials and Equipment

Video was taken with a Sony® Handycam DCR-TRV22 mini DV camcorder and edited on an Apple MacBook® laptop computer using the built-in iMovie® software program. The newer iMovie® program allows for cropping film clips, which allowed the team to remove adults from the movie when they came into camera range. Other features of iMovie® that were used in the production of the self-modeling video were titles; transitions between clips; and audio insertions, including clapping, music, and voiceovers. Sound effects and ambient music come with the software, so applying these to the video was simply a matter of clicking and dragging the content. (The same formatting is used for the PC software counterpart, Movie Maker®.)

The filming sessions on the playground utilized the material and equipment that were always present. Along with the larger playground equipment already mentioned, balls of various sizes, digging toys for sand play, and tricycles and small enclosed plastic cars were used in the videos and were available for use during recess.

Dependent Variables

Social initiations were targeted because of these children's lack of response to other methods, including teacher prompting, Social Stories™, and the buddy system. Physical and vocal social initiations with peers in a natural environment (playground at recess) were selected as the dependent variables. Operationalizing complex human interaction is difficult at best, and this was especially true with physical initiations in this study. Field-testing a data collection form for more than a month shaped the definitions used in this study. The goal was to keep the dependent variables true to the intent of the study while also ensuring that observers could agree on what they were seeing. Eight variations of the original form were made before the final version was adopted (data collection form available upon request). *Social initiation* was divided into *physical* and *vocal* modes. *Physical*

initiation was defined as moving into proximity of a peer or group of peers, resulting in some form of engagement. *Engagement* was defined in terms of attending to the peer or the activity of the peer (or peers) for more than 5 s while within arm's reach or actually making physical contact with the peer. *Vocal initiation* was defined as either *verbal* (use of words) or *nonverbal* (vocalization including laughing, crying, and utterances to gain attention). Because most vocal initiations required physical proximity, it was decided that vocal initiation would override physical initiation in the collection procedure in order to avoid duplication of data (i.e., one initiation counting as both physical and vocal). Thus, once a vocalization occurred, no physical initiation was recorded for that specific event. It was necessary to define the duration of an initiation so that repetitive activities did not carry undue weight. Once a physical initiation was carried out, no additional physical initiation was recorded unless the children disengaged for a period of 10 s or moved to another activity and then returned. For example, if a child engaged in playing catch and repeatedly threw and caught a ball, only the first action was counted. Repetitive vocal events were similarly not scored except for the first event; however, verbalizations (actual words or phrases) that were separated by at least 5 s and were not a direct result of verbal prompts of others were noted and counted as separate data points.

Observation and Data Collection

Four observers were used in this study: the primary author, who is a university professor; two graduate assistants; and an undergraduate honors student in physical therapy. Observers carried clipboards and moved into proximity of the child being observed so that verbalizations could be heard. When two observers were watching the same child for interobserver agreement purposes, they positioned themselves opposite of each other to the point allowed by the playground equipment. Two of the observers had spent considerable time on the playground (daily for approximately 5 months), carrying out other observations so that most of the children were used to them. The other two observers spent a month on the playground prior to collecting baseline data. Much of this time was spent observing and establishing interobserver agreement; however, the observers did take time to help out on the playground between observations and would respond to children during data collection. However, responses to students who were being observed were kept as brief as possible, and these were extremely rare.

The four-member research team collected frequency data during daily 15-min observations of participants during playground time. Each child was observed at least one time per day; occasionally two observations occurred on the same

day by independent observers. A chart was kept of the number of observations per child, as well as the name of the observer. A researcher did not observe a child two times on the same day. If any of the children fell behind the others in number of observations (due mainly to absenteeism), this child was observed more frequently until equity was restored. The order of viewing the children was dependent on when the classrooms came out for their playground time. The time was set for 9:30 each morning, but there was considerable variance. This in itself established a form of random selection, as the first two participants on the playground were each assigned to one of the observers and were the first to be observed. The other two observers either waited for the second pair to appear or participated in simultaneous observations for interobserver agreement.

Independent Variable

The independent variable in this study was a 2.5- to 3.5-min video featuring each participant socially interacting with peers. This included footage of the child initiating interactions and the result of initiations: playing together. Two methods were used to obtain sufficient footage. First, the child with ASD was taken to the playground with either one or two peers at times when no other children were present. The peers were prompted simply to try to get the child with ASD to interact with them. Planning was limited to the research team discussing particular behaviors that team members thought were within the child's ability and would be particularly important for the child's social development. Behaviors such as holding hands during the transition to the playground, tapping a child on the shoulder, throwing a ball to a peer, sharing a piece of equipment or toy, laughing, and so forth were encouraged once the filming session began. However, the researchers mainly followed the lead of the children, giving prompts to the peers at key times during play. For example, a peer was doing a parallel play activity as both he and Lucy played in the sand. The peer was prompted to share some digging tools with her, resulting in excellent footage of the exchange. Five peers were involved in these sessions, and they enjoyed the extra time on the playground and were very cooperative. Most of the video footage was captured in this way.

A small amount of footage came from having the observers take the camcorder as they collected data during baseline. Occasionally a situation would arise on the playground that could fit into a video. For example, footage was captured of a child throwing a ball with peers present. Even though the child was not throwing the ball to a peer, footage of the peer catching a ball could be taken. Combining these two clips made it appear that the child with ASD was playing catch with a peer. The ease with which this footage was collected and the short amount of time needed should be emphasized. The average amount of raw footage taken of each child was 15 min.

Graduate assistants assigned to the project (authors two and three) created the videos for the four participants. Clips were edited and arranged to make it appear as if the child with ASD was initiating interactions and playing with peers in an appropriate and fun manner. A flattering still-frame image of the child was selected and placed at the beginning of the movie, and one of the graduate students verbally labeled the behavior (e.g., "This is Lucy's Movie! Let's watch Lucy playing nicely with her friends."). Even though the children could not read, titles also were placed on the still frame to better resemble actual TV fare. Likewise, at the end of the movie another flattering still frame was placed, and "Good job playing, Lucy" was recorded over it. An audio clip of children cheering followed this. The use of a music background was done to eliminate background noise and adult prompts when present. When the child and peers were talking or laughing, the music was turned off and the actual sound was turned back up.

The videos were very similar in content and format, although certain behaviors depicted on each varied based on children's play preferences on the playground. Helen's video contained footage of her on a three-person tire swing because that is where she spent most of her time on the playground. Lucy spent more time than the other three playing in the sand, so footage was shot there. John had a fascination with balls, so scenes were constructed to make it appear he was sharing them and playing catch. When possible, language use on the playground was captured by having the child imitate while the researcher was taking close-up shots of the child's head and face. Tim and Lucy were able to imitate several words and simple phrases such as "c'mon," "let's play," "slide," and "ball." Video of these utterances was placed just prior to the demonstration of a matching behavior. John and Helen were not able to imitate, thus their videos had no examples of them verbalizing. To depict the idea of initiating play on John and Helen's video, a physical approach was used. Video was taken of them in proximity to peers and walking up to peers. Helen even touched some peers in passing. These approaches to peers were included in the video, followed by a scene depicting them at play with the peer who was approached. All of the videos had short vignettes of the participants transitioning from the classroom to the playground, as well as footage of the children with peers on the large pieces of equipment.

Tim's video turned out to be 3.5 min in length, whereas the other three had videos that were 2.5 to 3 min long. Dowrick (1983) stated that the optimum length of VSM videos is 2.5 min, with times greater than this not producing any differences in effect. The researchers monitored the children as they viewed their videos, and all four were able to attend to the screen for the entire session and all four exhibited behavior indicating enjoyment at watching themselves, including clapping, laughing, pointing, and bouncing.

Once the first video was created, and it was clear that the filming process had not changed baseline rates of behavior, Lucy began to view her video. One of the researchers went into the class with a laptop and took the participant to a quiet area (either the teacher's small office or an observation room for the classroom). The viewing occurred approximately 1 hr prior to recess. Lucy continued to view the video for 2 weeks, and then it was withdrawn. Tim began viewing his video shortly after Lucy's was withdrawn. This pattern continued until all four students had viewed their videos for a period of 2 weeks and were subsequently being evaluated for maintenance of new skills. The 2-week intervention phase was chosen for several reasons. In previous studies with preschoolers (Buggey, 2005; Hepting & Goldstein, 1996), the original videos shown to children needed to be revised before change was seen. In the present study, time was built into the intervention phase to allow for determining if change was occurring and to analyze videos to see if content needed to be amended if change was not evident; content was not modified in this study. The children moved directly into maintenance if a strong, positive, and consistent change was seen in behavior. The children watched their videos a total of 10 times during the 2-week intervention period, except Helen, who watched 8 times due to absences. The number of observations during intervention varied somewhat mainly due to absenteeism and conducting second observations in a single day. Lucy was observed 12 times, Tim 11 times, John 10 times, and Helen 8 times.

Design

The present research was conducted using a single-subject, multiple-baseline design across four students. Three methods were used to analyze data. The first was visual inspection of graphs, the second was examining differences in means across study phases, and the third was a calculation of percentage of nonoverlapping data (PND; Scruggs & Mastropieri, 1998). PND was used to augment the more traditional forms of data analyses.

The study was implemented from January to early June, with some of the interobserver agreement work occurring prior to the winter holiday in late December. Because four data collectors were involved, and due to the length of the study, an unusually large amount of data was obtained, allowing for well-established baselines and the assessment of relatively long periods of maintenance for three of the children. Bellini et al. (2007) noted in their study that many of the studies of VSM use with children with ASD had limited assessment of maintenance.

Reliability

Ninety percent agreement across all observers was set as the criterion for acceptable interobserver agreement. One

Table 2. Interobserver Agreement Totals Among Observer Pairs

	Observer 1	Observer 2	Observer 3
Observer 1			
Observer 2	97.6		
Observer 3	94.0	92.8	
Observer 4	91.5	90.3	92.0

month prior to beginning the baseline phase of the study, the interobserver agreement process began. The four observers collected data on children during playtimes and through video-recorded sessions that were viewed independently. Because the children with ASD rarely interacted, data were collected on typically developing peers or other children with developmental disabilities during many of the observation sessions. The observers then met as a team to review their findings and discuss differences of opinion. A grid was arranged so that agreement could be calculated across each of the six possible pairings of the observation team. Results were compared among the observers so that problems with definitions and the data collection form were addressed and the form refined as needed. Pairs who had less-than-acceptable agreement did additional observations until criterion was reached.

Several revisions of the data collection form and dependent variable definitions were made before the team felt comfortable that 90% agreement could be achieved. Once the form was agreed to, the observers continued to collect and compare data until the 90% agreement across the six possible pairs was established. To ensure the maintenance of the interobserver agreement, an additional 25% of observation sessions were completed with another team member throughout all study phases. Identical data collection procedures were used in the baseline, intervention, and maintenance phases of the study. Agreement above 90% was maintained throughout the intervention and maintenance sessions. A summary of the totals of agreement across the observers is presented in Table 2.

Results

Three of the four participants in this study made gains in the frequency of their social initiations based on visual inspection of the data. There was no effect seen for John. A summary of the mean number of initiations across phases of the study is illustrated in Table 3. The results per data collection session are shown in Figure 1.

Lucy increased her mean number of social initiations during intervention as compared to baseline, and the mean number in the maintenance phase rose further. The type of initiations changed markedly for Lucy. During baseline, only 3 of her initiations (23%) were vocal (all laughter), whereas the proportion during intervention rose to 42% of her total

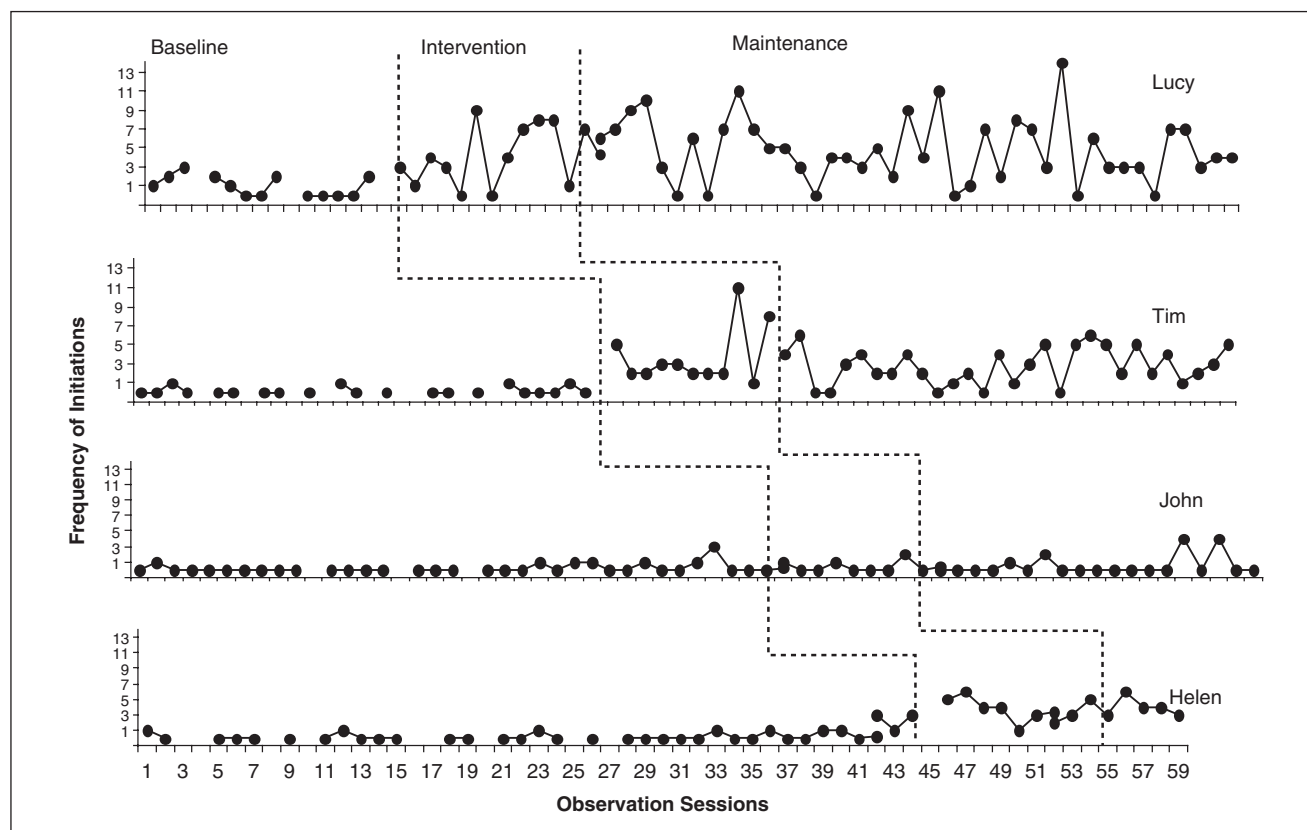


Figure 1. Number of social initiations during 15-min sessions

Table 3. Mean Number of Initiations per 15-min Observation Session

Student	Baseline M	Intervention M	Maintenance M
Lucy	1.14	4.33	4.83
Tim	0.19	3.73	2.77
John	0.26	0.44	0.58
Helen	0.22	3.33	3.75

initiations or 1.83 vocalizations per observation session, approximately half of which were verbal. This trend continued throughout the maintenance phase, where her mean number of vocal initiations rose to 2.55 (48%) per session.

Tim's mean number of social initiations rose during intervention and decreased during maintenance (but remained greater than those seen in the baseline phase). Of note is the change in the number of vocal initiations to peers. Tim was never heard verbalizing to peers but made nonverbal utterances (15%) across the 16 baseline sessions. During the 10 sessions of the intervention phase, he communicated vocally 19 times (14% of his social initiations), 5 of which involved words.

John exhibited no changes in behavior. What appear to be gains in the maintenance phase represent John approaching other children who picked up balls he rolled down slides. In

fact, reacting to children who took balls accounted for every initiation John exhibited in the study.

Helen exhibited a substantial increase in mean number of socializations during intervention and maintenance. Helen only initiated vocally twice during baseline (1% of her social initiations), and these were nonverbal utterances aimed at getting a peer to leave her alone. During the intervention phase, Helen vocally communicated to peers eight times (41% of her social initiations).

Percentage of Nonoverlapping Data

PND also was used to analyze results in this study. PND was calculated by counting the number of data points in the treatment or maintenance phase that exceed the highest baseline data point and dividing by the total number of data points in the phase and multiplying by 100. Lucy's PND scores of 58% in intervention and 53% in maintenance are indicative of questionable results (Scruggs & Mastropieri, 1998). Tim's PND scores were 91% for treatment and 77% for maintenance, both indicating effective treatment. Calculation of Helen's PND indicated scores of 78% in treatment and 100% in maintenance. Both scores represented strong treatment effects.

Social Validity

Information was gathered from adults concerning the VSM procedure, its application in the classrooms, its effects on the students, and the feasibility of using VSM in the future. Information was procured through informal interviews beginning one week into the first intervention phase and continuing once every other week for the remainder of the study. Teachers, therapists, and parents were asked the following questions: “Do you believe VSM is helping the child?” “Does the implementation cause any disruption to the classroom routine?” “Does the child seem to [still] enjoy watching the videos?” “Is the implementation distracting to other students?” and (for parents) “Does your child communicate to you about being in a movie?” More formal assessment of social validity was not undertaken because the researchers did not want the teachers to associate extra paperwork with the method, and it was felt that more accurate information could be gained from conversations, most of which were carried out on the playground or when parents were picking up their children. At the end of the study when the procedure for making VSM videos was explained to the adults, they also were asked about the practicality of implementing VSM on their own in the future.

The two teachers were universally supportive of the method, especially after positive results were noticed. They reported almost no disruption to class routines and reported that the children loved to watch their videos. One child would sign for “movie” at other times during the day. However, the reports of treatment efficacy differed between the teachers. The teacher of John and Helen was positive but reserved. Lucy and Tim’s teacher was extremely positive and delighted with the method. This makes sense given that John made no progress and Helen’s changes seemed more subtle than Lucy’s and Tim’s. Both teachers voiced other possible applications of VSM as the study progressed. In terms of practicality, the therapists and parents were more enthusiastic about actually making their own VSM videos. The teachers stated they had little time to devote to filming and editing. The therapists and parents, on the other hand, thought that the time necessary to produce self-modeling videos was available in their schedules, and the therapists were excited about adding this to their repertoires of interventions.

Discussion

The implementation of a VSM intervention appeared to be successful across three of four study participants, although calculations using PND made claims of success problematic for Lucy. Anecdotal information indicated unanticipated effects across other playground behaviors that served to support positive outcomes. The teachers and therapists, as well as the observers, were asked to note any changes in

behavior. There were several changes in behavior that were reported by preschool staff, parents, and observers that were only obliquely related to the target behaviors. Compared to baseline, Tim left his “zone of comfort” in his little car and started driving the length of the playground. He spent less time in the car and occasionally did not use it at all during playground time. Instead, he began to use playground equipment such as the slide and swings, something that did not occur in baseline. His movie contained a scene showing him sliding side-by-side with a peer. In reality, he did not complete that slide but was shown together with the peer at the top of the slide and then the scene shifted to the legs of two peers coming down the slide. This may have influenced his willingness to slide during intervention. Tim also began approaching adults and verbalizing to them. His teacher jokingly stated several times, “Who is this child?” indicating a broad-based change in behaviors. Tim started approaching his peers physically and began participating in activities but made few verbal contacts with them (although when he did these tended to be actual verbalizations). He spent time hovering near groups of peers and following them. An interesting behavior that arose shortly after viewing his video was standing in front of large tinted picture windows facing the playground, and engaging in verbalizations and play with his reflection.

Anecdotally, the most striking change in Lucy reported by her teachers and therapists (and reports from home) was her frequency of vocalizations. During baseline, Lucy had four vocal initiations during the 3.5 hrs of observation, all involving laughter. Following intervention, she began to verbalize on the playground, including calling two of her peers by name. Lucy’s graph (see Figure 1) shows great variations in the numbers of initiations, including several days without initiations during intervention. This reflects that although Lucy engaged more with peers, she did not abandon her play with plants. This is very understandable considering that the purpose of this study was not to eliminate the ritualistic or habitual behaviors of the participants. Rather, it was to promote new behaviors. Lucy’s fascination with foliage and greenery did not stop; thus, if Lucy happened to be selected for observation on a day when dandelions were blooming, she could easily spend the whole 15 min manipulating flowers. The scores she obtained in the maintenance phase of 14 initiations followed by 0 were obtained on the same day. Thus, any of the evaluation measures that use comparisons based on lowest scores between phases would be problematic for Lucy. It would seem that visual inspection and examination of means might be the most appropriate form of evaluation for Lucy. It could be argued that the PND score actually supports claims for improvement with Lucy. Although the mean changes for Lucy’s and Tim’s behaviors were fairly comparable, the mean number of initiations for Lucy when not engaged in her plant activities was greater than Tim’s.

John exhibited no changes in behavior at any time during the study. All occasions when John did interact with others involved peers taking balls with which he was playing. These initiations involved trying to retrieve the balls, and all involved physical proximity or vocal initiations (non-verbal). The fact that the two scores in maintenance were elevated was more related to the persistence of the peer in not relinquishing the balls than in John being more social. The first author of this article has worked with 30 students using VSM, including 15 who had autism. The only time VSM did not produce results was when the child exhibited difficulty attending to the video. This was not the case with John. He avidly watched his video and usually clapped at the end.

Helen had an upswing in initiations after she began to watch her video, including reaching out and taking the hand of a peer. The accompanying changes in other behaviors noted with Lucy and Tim were not as pronounced with Helen, although observers and teachers noticed subtle changes on the playground, including increased interest in group activities and approaching adults.

Limitations

A constant threat to validity in single-subject design is the small sample size. In this study, that threat was confounded by John's lack of change. However, three of the four participants exhibited changes that aligned with the introduction of the VSM videos. Measures of social validity involving teacher and therapist attribution of these changes to watching the videos added some strength to the results.

Another threat that is particularly worrisome when working with preschool-age children is maturity. This threat was somewhat neutralized by the extended baselines and being able to use as a referent information about social interactions from the study that took place during the previous school year.

There also was a threat of contamination in this study. Although students watched their tapes in private, the filming was often done with the knowledge (and sometimes the participation) of other students. Also, the presence of the researchers in all observation sessions may have served as a prompt for both participant and peer to engage in social behaviors. Although there was much effort dedicated to desensitizing the students to the researchers' presence, this cannot be eliminated as a possible confound.

The density of children on the playground probably affected results. At various times there could be from one to four classes on the playground. This meant that between 12 and 48 children could be present. The opportunities for interactions would increase as the population grew, and likewise the opportunity to avoid interactions with and proximity to others decreased.

One area that might be described as both a strength and weakness of this study was the analysis of maintenance. Although there were relatively frequent observations during maintenance phases for Lucy and Tim, there were fewer opportunities for John and Helen, due to the school year coming to an end.

A final threat to validity was absences due to illnesses. All of the children were absent for some of the time during the study. Tim, John, and Helen had prolonged absences of more than one week.

Conclusions and Suggestions for Further Research

The findings of the present study suggest positive outcomes of using VSM to stimulate social initiations of preschoolers with ASD, although the failure of John to make progress raises questions. Still, the limited number of effective strategies available to parents and professionals for working with this population should result in VSM receiving more attention as a viable method. One of the criticisms often directed at VSM (as seen in critiques on article submissions and grant proposals) involves the technology requirements. Some believe that video editing and filming take too long and that much of the process is beyond the abilities of potential users. This may have been true in the past but is no longer the case. With a little planning, 3-min videos can be produced in less than an hour. The skills necessary to complete the editing can be learned in 15 min and in some ways are less complicated than word-processing skills. This can be illustrated by the ease with which the graduate students learned to do the editing. The three graduate assistants were trained by the first author for approximately a half hour on iMovieHD® (a similar, but simpler program to use than iMovie®). The next day, the graduate students were producing videos with overlaid sound on iMovie® and were instructing the first author on the differences in the two programs.

As the positive effects of using videos in all forms to train children with ASD are receiving growing attention, it is important to factor VSM into the equation. In fact, in several of the studies on VSM, including the present study, children exhibited significantly improved behavior when more conventional methods had proven ineffective (e.g., Bellini et al., 2007; Buggey, 2005).

John's lack of progress leads to several questions that should receive additional attention by researchers. John was the youngest of the participants in this study. There must be limits to the population that can benefit from VSM, involving chronological age, cognitive ability, social maturity, and other variables. These factors need to be explored thoroughly to better determine who will benefit most from VSM.

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